R3 v4 — ASI Autonomous Engineering System (v2)

Fully aligned with PRD v4.0. Defines a self-healing, triple-validated, AI-native development system integrated into the R3 monorepo and LLPTE architecture.

Core Principle

Trust Nothing. Verify Everything x3. Every code change must pass three independent validation layers before commit.

System Architecture

Four agents: Orchestrator (controls flow), Auditor (analyzes repo + PRD rules), Refactor Agent (generates scoped patches), Validator (runs triple-check pipeline).

Agent Graph (ASI Layer)

Intent → Orchestrator → Auditor → Refactor → Validator → (Pass x3 → Commit) or (Fail → Self-Heal Loop).

Triple Validation Pipeline

1. Static: TypeScript strict, ESLint, dependency graph. 2. Runtime: isolated execution + component rendering. 3. Regression: compare behavior vs baseline.

Self-Healing Loop

On failure: rollback → diagnose → regenerate fix → re-run validation. Continues until safe or escalated.

PRD Hard Guard Enforcement

No any, no console.log, no unsafe writes, no broken routing, strict type guarantees, full dependency awareness before modification.

aiDecisionLog Integration

All AI actions logged: confidence, latency, outcome. Feeds system learning loop and acceptance tracking.

LLPTE Integration

Agent system mirrors LLPTE pipeline structure. Code decisions treated as 'engineering inference' with measurable latency and confidence.

Monorepo Integration

Hooks into pnpm workspace. Scans client/, server/, packages/. Prevents phantom directory usage and invalid imports.

Bug Resistance Model

Prevention (analysis), Detection (validation), Correction (self-heal). No change reaches production without triple pass.

Outcome

Autonomous, self-evolving, bug-resistant engineering system capable of maintaining R3 v4 at near-zero defect state.

Next Evolution (ASI Path)

Predictive bug modeling, architecture auto-rewrites, confidence scoring for code changes, and continuous repo optimization.